

# Cardiac Arrest at Exercise Facilities

## Implications for Placement of Automated External Defibrillators

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### Objectives

This study sought to characterize the relative frequency, care, and survival of sudden cardiac arrest in traditional indoor exercise facilities, alternative indoor exercise sites, and other indoor sites.

### Background

Little is known about the relative frequency of sudden cardiac arrest at traditional indoor exercise facilities versus other indoor locations where people engage in exercise or about the survival at these sites in comparison with other indoor locations.

### Methods

We examined every public indoor sudden cardiac arrest in Seattle and King County from 1996 to 2008 and categorized each event as occurring at a traditional exercise center, an alternative exercise site, or a public indoor location not used for exercise. Arrests were further defined by the classification of the site, activity performed, demographics, characteristics of treatment, and survival. For some location types, annualized site incident rates of cardiac arrests were calculated.

### Results

We analyzed 849 arrests, with 52 at traditional centers, 84 at alternative exercise sites, and 713 at sites not associated with exercise. The site incident rates of arrests at indoor tennis facilities, indoor ice arenas, and bowling alleys were higher than at traditional fitness centers. Survival to hospital discharge was greater at exercise sites (56% at traditional and 45% at alternative) than at other public indoor locations (34%;  $p = 0.001$ ).

### Conclusions

We observed a higher rate of cardiac arrests at some alternative exercise facilities than at traditional exercise sites. Survival was higher at exercise sites than at nonexercise indoor sites. These data have important implications for automated external defibrillator placement. (J Am Coll Cardiol 2013;62:2102–9) © 2013 by the American College of Cardiology Foundation

Regular exercise can reduce an individual's overall risk of sudden cardiac arrest (SCA). However, exercise transiently increases one's SCA risk during and immediately following activity among those who do and do not exercise regularly (1,2). This association has prompted both voluntary and compulsory placement of automated external defibrillators (AEDs) in exercise facilities, although AED placement is not universal and requirements are inconsistent from state to state (3,4). In addition, there are few data available regarding the utility of placing AEDs at exercise facilities.

Site incidence rates of SCA at public locations, including traditional health and fitness centers, have been explored

previously (5). However, little is known about the frequency of SCA at other indoor locations where some form of exercise occurs, such as bowling alleys, dance studios, and church gyms or community centers or whether better preparedness for such events in these places, including the ready availability of an AED, would be beneficial.

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The purpose of this study was to characterize the frequency, treatment, and outcome of SCA occurring at both traditional and alternative exercise facilities. Through such study, we sought to better inform the selection of sites for possible AED placement.

### Methods

**Study design and population.** This was a retrospective cohort study composed of persons with nontraumatic cardiac arrest that occurred in a public indoor location within Seattle and King County between January 1, 1996, and December 31, 2008, who were treated by emergency medical services

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(EMS) personnel. Patients of all ages whose arrests occurred prior to EMS arrival were included. Cardiac arrest cases were classified into 3 groups based on indoor location: traditional exercise facilities, alternative exercise facilities, and nonexercise facilities.

**Definition of exercise facility.** There is no standardized definition of an exercise facility. Accordingly, we developed a broad definition to include any public indoor location where exertional activities are performed for physical training and/or recreation. We further divided exercise facilities into traditional and alternative exercise facilities. Traditional exercise facilities included health clubs and fitness centers—commercial exercise sites specifically intended for physical training. Alternative exercise facilities included locations that might be regarded as more informal or recreational in their exercise focus, such as bowling alleys, workplace or hotel gyms, church or civic gyms, and dance studios. All other public indoor locations, such as banks, restaurants, shopping centers, airports, etc., were categorized as nonexercise facilities.

**Data collection and definitions.** The Seattle Fire Department has maintained a registry of all cardiac arrests receiving pre-hospital emergency medical treatment in the city of Seattle since 1970. King County EMS has maintained a similar registry of treated out-of-hospital cardiac arrests outside the city of Seattle since 1976. The registries collect information and code data elements according to the Utstein definitions (6,7). These include outcome measures such as return of spontaneous circulation (ROSC), hospital admission, and survival to hospital discharge.

EMS responders in Seattle and King County complete a pre-hospital report for every patient experiencing cardiac arrest for whom they provide care, documenting the details about the location, circumstances, and treatment provided. We reviewed the incident report forms for each of the public indoor arrest events to determine the specific arrest location. When the type of facility could not be ascertained from the forms, we employed Web-based sources such as <http://maps.google.com>, <http://www.google.com>, <http://www.bing.com>, and <http://www.yellowpages.com> to characterize the location. Location type was determined in conjunction with the arrest date because business type can change over time. Data regarding the number of participants at each facility were not available; therefore, no estimate of the per-person risk could be calculated. Bystander cardiopulmonary resuscitation (CPR) was defined as administration by persons at the scene, prior to arrival of EMS; no data are available on the level of training of those providing CPR.

We also collected information about the specific type of activity in which the patient was engaged. We used this information to classify the type of physical activity according to the Bethesda Conference sports matrix (8). This classification grades both the static component of exercise as low (I), moderate (II), or high (III) and the dynamic component as low (A), moderate (B), or high (C). Information about the timing of the arrest in relation to the

activity and other arrest characteristics were abstracted from the incident report forms for arrests that occurred in exercise facilities. The exercise performed at the time of arrest was not presumed based simply on location, even for bowling alleys.

We utilized Walls and Associates' time-series database of business establishments, the National Establishment Time-Series (NETS) Database (9), to calculate a denominator for some of the types of exercise facilities. We counted the number of establishments listed in the NETS Database during 2002, the midpoint of our case series, to calculate an

annualized site incidence rate of cardiac arrest. The number of community centers and hotels with gyms could not be ascertained from the NETS Database; therefore, we conducted Web-based searches to estimate the number of sites. The study was approved with exemption from informed consent by the University of Washington Human Subjects Division.

**Statistical analysis.** We conducted descriptive analyses to characterize our study population and the categorized location of their cardiac arrest. For all further analyses, the study population was restricted to those whose presenting cardiac arrest rhythm was ventricular fibrillation or ventricular tachycardia (VF/VT). To identify differences across all 3 location groups, we first conducted 3-way comparison analyses, using Pearson chi-square tests and 1-way analysis of variance. Then, independent sample Student *t* tests were used to compare mean age and mean EMS response times between groups. Differences between categorical variables were assessed using Pearson chi-square tests and Fisher exact tests when the expected cell counts were <5. We used multivariate analysis to control for potential confounders by fitting the data to a binary logistic regression model to compare survival between exercise and nonexercise facility cases. Due to the number of comparisons, we used a *p* value of 0.01 to determine statistical significance. Statistical analyses were conducted using IBM SPSS version 20.0 (Chicago, Illinois).

## Results

**Study population.** Between January 1, 1996 and December 31, 2008, 865 episodes of nontraumatic SCA occurred in a public indoor location. In 16 cases, the location of arrest was not available. Of the remaining 849 arrests, 136 occurred in an exercise facility—with 52 at traditional fitness or health clubs and 84 at alternative exercise sites. The number of SCAs and their location are shown in Table 1.

### Abbreviations and Acronyms

|                                                                 |
|-----------------------------------------------------------------|
| <b>AED</b> = automated external defibrillator                   |
| <b>CPR</b> = cardiopulmonary resuscitation                      |
| <b>EMS</b> = emergency medical services                         |
| <b>NETS</b> = National Establishment Time-Series                |
| <b>PAD</b> = public access defibrillation                       |
| <b>ROSC</b> = return of spontaneous circulation                 |
| <b>SCA</b> = sudden cardiac arrest                              |
| <b>VF/VT</b> = ventricular fibrillation/ventricular tachycardia |

**Table 1** Public Indoor Locations Defined as Typical and Alternative Exercise Facilities

| Traditional Exercise Facilities | Number of Cardiac Arrests | Nontraditional Exercise Facilities  | Number of Cardiac Arrests |
|---------------------------------|---------------------------|-------------------------------------|---------------------------|
| Health clubs/fitness centers    | 52                        | Basketball courts in churches       | 1                         |
|                                 |                           | Basketball courts in juvenile halls | 1                         |
|                                 |                           | Bowling alleys                      | 11                        |
|                                 |                           | Community centers                   | 33                        |
|                                 |                           | Dance studios                       | 4                         |
|                                 |                           | Golf course clubhouses              | 2                         |
|                                 |                           | Hotel gyms                          | 4                         |
|                                 |                           | Ice arenas                          | 3                         |
|                                 |                           | Indoor paintball centers            | 1                         |
|                                 |                           | Martial arts schools                | 1                         |
|                                 |                           | Recreation area in senior centers   | 7                         |
|                                 |                           | Roller skating rinks                | 1                         |
|                                 |                           | School gyms                         | 4                         |
|                                 |                           | Ski school clubhouse                | 1                         |
|                                 |                           | Tennis facilities                   | 6                         |
|                                 |                           | Workplace gyms                      | 4                         |

The clinical and resuscitation characteristics of patients are shown in Table 2. Patients, the majority of whom were male, averaged  $58.6 \pm 16.4$  years of age. Male sex was more common among individuals experiencing arrest in exercise facilities than in indoor public locations where exercise was not occurring (91% vs. 76%;  $p < 0.001$ ). Patients with exercise-associated SCA were also younger than those in whom SCA was not associated with exercise (mean age 55.1 vs. 59.2 years, respectively;  $p = 0.007$ ).

**Site incidence rates of cardiac arrests.** The annualized site incidence rate of SCA at health clubs and traditional fitness centers was 0.024 arrests/site each year (Table 3). This translates to 1 arrest every 42 years at each traditional exercise facility site. We observed a higher rate at indoor tennis facilities, ice arenas, and bowling alleys than at traditional exercise facilities, with one arrest every 11, 13, and 27 years/site, respectively. The incidence rate at community centers, 1 arrest every 51 years/site, was similar to that of traditional exercise facilities. Other alternative exercise facilities had lower site incidence rates of SCA.

**Exercise activities associated with SCA.** Documentation of the specific type of exercise associated with SCA was available in 112 cases (Table 4). Of the remaining 24 cases, 5 patients were documented not to have been performing any exercise at the time of their arrest and the precise activity was not documented in 19 patients. The most frequent activities included basketball (20.5%), dancing (11.6%), “working out” (11.6%), treadmill (8.9%), tennis (6.3%), bowling (5.4%), and swimming (4.5%). The temporal relationship between cardiac arrest and exercise was documented

in 119 cases; among these, the arrest occurred during exercise in 92 (77%), after exercise in 22 (18%), and before (or unrelated to) exercise in 5 (4%). In no case was there a report of a strike to the chest that would suggest commotio cordis, a cause of SCA associated with sports that has been shown to respond to early defibrillation (10).

When classified according to the Bethesda Conference sports matrix (8), the exercise types were distributed, from more to less frequent, as IIC (moderate static and high dynamic components) in 32.1%, IC (low static and high dynamic components) in 25%, IB (low static and moderate dynamic components) in 14.3%, and IIB (moderate static and moderate dynamic components) in 14.3%. Four times as many patients (58%) were participating in high dynamic exercises (IC, IIC, and IIC) than in low dynamic exercises (13.4% with IA, IIA, and IIIA) at the time of SCA.

**Resuscitation outcomes.** As shown in Table 2, arrests in exercise facilities (traditional and alternative) were more likely to be witnessed ( $p < 0.001$ ) and of cardiac etiology ( $p < 0.001$ ), present with VF/VT as the initial arrest rhythm ( $p < 0.001$ ), and receive bystander CPR ( $p < 0.001$ ) and public access defibrillation (PAD) ( $p = 0.003$ ). In addition, bystander CPR and PAD application were provided to more traditional exercise facility arrests (83% and 25%, respectively) than to alternative and nonexercise facility arrests combined (64% and 9%;  $p = 0.007$  and  $p < 0.001$ , respectively).

Considering univariate outcome endpoints—ROSC, hospital admission, and survival to hospital discharge—only survival was statistically significantly different among the 3 location categories. Survival was higher in exercise facilities than in nonexercise facilities (49% vs. 34%;  $p = 0.001$ ). Survival was also higher ( $p = 0.003$ ) when we compared traditional exercise facilities (56%) with alternative and nonexercise facilities (38%). There was no difference in survival to discharge when SCA occurred at a traditional as compared with alternative exercise location ( $p = 0.233$ ).

**Patients with initial VF/VT.** VF/VT was the initial rhythm documented in 113 of the 136 patients (83.1%) at typical or alternative exercise facilities. The other initial rhythms among patients at these locations were pulseless electrical activity ( $n = 10$  [7.4%]), asystole ( $n = 10$  [7.4%]), bradycardia ( $n = 2$  [1.5%]), and sinus tachycardia ( $n = 1$  [0.7%]). A shockable initial arrest rhythm accounted for 65% of all 849 SCAs at all locations under study. The characteristics of these patients are shown in Table 5.

Among patients experiencing VF/VT, only age and sex were statistically significantly different across all 3 location groups (Table 5). Patients in exercise facilities were younger than patients in nonexercise facilities ( $p < 0.001$ ). Male sex was more common among those with cardiac arrest in exercise facilities than nonexercise facilities ( $p < 0.001$ ). Comparing exercise facilities with nonexercise facilities, the frequency of having a witnessed arrest was similar between the 2 locations; however, bystander CPR seemed more frequently initiated and PAD more likely to be applied

**Table 2** Data Analysis for All Public Indoor Location Cardiac Arrests That Occurred Prior to EMS Arrival\*

| Variable                | Total Included Arrests (n = 849) | Traditional (T) Exercise Facility Arrests (n = 52) | Alternative (A) Exercise Facility Arrests (n = 84) | Nonexercise (N) Facility Public Indoor Arrests (n = 713) | Overall 3-Way Comparison Among T, A, and N Exercise Facility Arrests, Test Statistic and p Value | Activity-Based Comparison (T + A vs. N Exercise Facility Arrests), p Value | Traditional vs. Alternative Exercise Facilities, p Value | Location-Based Comparison (T vs. A + N Exercise Facility Arrests), p Value |
|-------------------------|----------------------------------|----------------------------------------------------|----------------------------------------------------|----------------------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------|
| Age, yrs                | 60.0, 58.6 ± 16.4                | 54.5, 55.1 ± 14.5                                  | 54.5, 55.1 ± 18.5                                  | 60.0, 59.2 ± 16.2                                        | 3.63<br>p = 0.027                                                                                |                                                                            |                                                          |                                                                            |
| Male                    | 669 (79)                         | 47 (90)                                            | 77 (92)                                            | 545 (76)                                                 | 14.88<br>p = 0.001                                                                               | <0.001                                                                     | 0.798                                                    | 0.035                                                                      |
| Witnessed               | 644 (76)<br>n = 843              | 46 (89)                                            | 76 (91)                                            | 522 (74)<br>n = 707                                      | 16.01<br>p < 0.001                                                                               | <0.001                                                                     | 0.707                                                    | 0.034                                                                      |
| Cardiac etiology        | 712 (84)<br>n = 848              | 50 (96)                                            | 79 (94)                                            | 583 (82)<br>n = 712                                      | 14.37<br>p = 0.001                                                                               | <0.001                                                                     | 0.708†                                                   | 0.010†                                                                     |
| Initial rhythm VF/VT    | 544 (65)<br>n = 843              | 41 (79)                                            | 72 (86)                                            | 431 (61)<br>n = 707                                      | 25.01<br>p < 0.001                                                                               | <0.001                                                                     | 0.299                                                    | 0.026                                                                      |
| Bystander CPR           | 551 (65)<br>n = 843              | 43 (83)                                            | 69 (82)                                            | 439 (62)<br>n = 707                                      | 20.68<br>p < 0.001                                                                               | <0.001                                                                     | 0.935                                                    | 0.007                                                                      |
| PAD applied             | 84 (10)                          | 13 (25)                                            | 10 (12)                                            | 61 (9)                                                   | 15.12<br>p = 0.001                                                                               | 0.003                                                                      | 0.048                                                    | <0.001                                                                     |
| EMS response, mean      | 4.5 (2.0)                        | 4.5 (1.9)                                          | 4.6 (2.1)                                          | 4.5 (2.0)                                                | 0.27<br>p = 0.763                                                                                |                                                                            |                                                          |                                                                            |
| ROSC at end of EMS care | 520 (61)                         | 35 (67)                                            | 48 (57)                                            | 437 (61)                                                 | 1.40<br>p = 0.496                                                                                |                                                                            |                                                          |                                                                            |
| Admitted to hospital    | 515 (65)<br>n = 795              | 35 (73)<br>n = 48                                  | 49 (62)<br>n = 79                                  | 431 (65)<br>n = 668                                      | 1.68<br>p = 0.433                                                                                |                                                                            |                                                          |                                                                            |
| Discharged alive        | 311 (37%)                        | 29 (56%)                                           | 38 (45%)                                           | 244 (34%)                                                | 12.67<br>p = 0.002                                                                               | 0.001                                                                      | 0.233                                                    | 0.003                                                                      |

Values are median, mean ± SD, or n (%), or min ± SD. \*n = 849 cases with known location type. †p value calculated using Fisher exact test.

CPR = cardiopulmonary resuscitation; EMS = emergency medical services; PAD = public access defibrillation; ROSC = return of spontaneous circulation; VF/VT = ventricular fibrillation/ventricular tachycardia.

**Table 3** Site Incidence Rates of Cardiac Arrests

| Exercise Facilities          | Number of Arrests | Number of Sites | Annualized Site Incidence Rate (Number of Arrests/Site/Year) | Number of Years to Have 1 Arrest/Site |
|------------------------------|-------------------|-----------------|--------------------------------------------------------------|---------------------------------------|
| Health clubs/fitness centers | 52                | 167             | 0.024                                                        | 42                                    |
| Indoor tennis facilities     | 6                 | 5               | 0.092                                                        | 11                                    |
| Ice arenas                   | 3                 | 3               | 0.077                                                        | 13                                    |
| Bowling alleys               | 11                | 23              | 0.037                                                        | 27                                    |
| Community centers            | 33                | 130             | 0.020                                                        | 51                                    |
| Roller skating rinks         | 1                 | 7               | 0.011                                                        | 91                                    |
| Dance studios                | 4                 | 165             | 0.002                                                        | 536                                   |
| Hotel gyms                   | 4                 | 225             | 0.001                                                        | 731                                   |
| Martial arts schools         | 1                 | 160             | <0.001                                                       | 2,080                                 |

at exercise sites, although these did not meet statistical significance. EMS response times did not significantly differ between exercise facilities and nonexercise facilities (4.6 vs. 4.4 min). There were no statistically significant differences in the univariate outcome measures—ROSC, hospital admission, and survival to hospital discharge—across the 3 location groups.

## Discussion

This study characterized cardiac arrests occurring at public indoor exercise facilities in a metropolitan community and demonstrated a substantial number of occurrences at both traditional and alternative exercise sites. We observed a higher site incidence rate at some alternative exercise facilities (e.g., tennis facilities and bowling alleys) than at traditional exercise facilities. More arrests occurred at alternative facilities than at traditional health and fitness centers. It is important to emphasize that these higher site incidence rates pertain to location and do not imply assessment of individual risk of participants of specific activities or at specific facilities. Overall survival was high and was greater at exercise facilities (whether traditional or alternative), as compared with other indoor public locations.

**Risk of SCA with exercise.** Regular physical exercise is associated with increased overall survival, although the risk of SCA is acutely increased at the time of exercise for both physically active and more sedentary individuals (1,2). Among patients with coronary artery disease, there is up to a 17-fold increased risk of SCA associated with 30 min of vigorous exertion as opposed to minimal exertion or rest (1). This relationship between acute risk of SCA and exercise, as well as reports of high frequency of arrest in fitness facilities (up to 3% to 6% likelihood of SCA per site per year) (4)

**Table 4** Activities at the Time of Cardiac Arrest

| Activity                      | Frequency | Percent | Exercise Classification |
|-------------------------------|-----------|---------|-------------------------|
| Aerobics                      | 2         | 1.8     | IIB                     |
| Basketball                    | 23        | 20.5    | IIC                     |
| Bowling                       | 6         | 5.4     | IA                      |
| Dancing                       | 13        | 11.6    | IB                      |
| Diving                        | 1         | 0.9     | IIA                     |
| Exercise cycle                | 1         | 0.9     | IIIC                    |
| Ice hockey                    | 3         | 2.7     | IIC                     |
| Martial arts                  | 2         | 1.8     | IIIA                    |
| Paintball                     | 1         | 0.9     | IC                      |
| Pickleball                    | 1         | 0.9     | IC                      |
| Playing in swimming pool      | 1         | 0.9     | IC                      |
| Playing pool                  | 1         | 0.9     | IA                      |
| Racquetball                   | 4         | 3.6     | IC                      |
| Refereeing basketball game    | 1         | 0.9     | IA                      |
| Roller skating                | 1         | 0.9     | IC                      |
| Running                       | 2         | 1.8     | IIC                     |
| Soccer                        | 1         | 0.9     | IC                      |
| Some kind of exercise program | 1         | 0.9     | IIB                     |
| Squash                        | 2         | 1.8     | IC                      |
| Swimming                      | 5         | 4.5     | IIC                     |
| Table tennis                  | 1         | 0.9     | IB                      |
| Tennis                        | 7         | 6.3     | IC                      |
| Treadmill                     | 10        | 8.9     | IC                      |
| Volleyball                    | 2         | 1.8     | IB                      |
| Water aerobics                | 3         | 2.7     | IIC                     |
| Weight lifting                | 4         | 3.6     | IIIA                    |
| Working out                   | 13        | 11.6    | IIB                     |
| Total*                        | 112       | 100.0   |                         |

Physical activity was defined using the Sports Classification Matrix developed at the 36th Bethesda Conference of the *Journal of the American College of Cardiology* (8). All activities were included and categorized based on the dynamic and static components of exercise defined by this matrix. \*This total excludes 5 patients whose arrests had no relation to exercise and 19 patients with unknown activity at the time of arrest.

prompted the American Heart Association and the American College of Sports Medicine to recommend AED placement in health and fitness facilities (11) and led to the development of various laws and regulations requiring preparedness for SCA, including training in CPR, development of emergency protocols, and placement of AEDs (4). However, these regulations are not consistent in terms of required protocols and characteristics of the locations that fall under the requirements (4); also, the implementation of recommendations is variable. For example, Drezner et al. (4) found that in health and fitness facilities in King County, Washington, only 40% had an AED on site; the true implementation of AED placement may well be lower in this cohort because only 63 of 136 facilities (46%) responded to the survey and there may be bias toward reporting placement of AEDs among the responders.

**Location of SCA at exercise facilities.** Becker et al. (5) categorized traditional health clubs/gyms as “higher-incidence” SCA public locations, with a site incidence rate of SCA of 1 per site every 12 years. We observed a lower site incidence in this geographically matched contemporary

**Table 5** Data Analysis for All VF/VT Arrests in Public Indoor Locations That Occurred Prior to EMS Arrival\*

|                         | Total Included Arrests (n = 544) | Traditional (T) Exercise Facility Arrests (n = 41) | Alternative (A) Exercise Facility Arrests (n = 72) | Nonexercise (N) Facility Public Indoor Arrests (n = 431) | 3-Way Comparison Among T, A, and N Exercise Facility Arrests, Test Statistic and p Value | Activity-Based Comparison (T + A vs. N Exercise Facility Arrests), p Value | Location- and Activity-Based Comparison (T vs. A Exercise Facility Arrests), p Value | Location-Based Comparison (T vs. A + N Exercise Facility Arrests), p Value |
|-------------------------|----------------------------------|----------------------------------------------------|----------------------------------------------------|----------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Age, yrs                | 62.0, 60.3 ± 14.9                | 54.0, 55.3 ± 13.7                                  | 54.0, 54.6 ± 18.1                                  | 62.0, 61.7 ± 14.1                                        | 8.12<br>p = 0.001 <sup>‡</sup>                                                           | <0.001                                                                     | 0.813                                                                                | 0.029                                                                      |
| Male                    | 441 (81)                         | 40 (98)                                            | 67 (93)                                            | 334 (78)                                                 | 17.67<br>p < 0.001                                                                       | <0.001                                                                     | 0.414 <sup>‡</sup>                                                                   | 0.003 <sup>‡</sup>                                                         |
| Witnessed               | 497 (92)<br>n = 541              | 39 (95)                                            | 68 (94)                                            | 390 (91)<br>n = 428                                      | 1.54<br>p = 0.463                                                                        |                                                                            |                                                                                      |                                                                            |
| Cardiac etiology        | 533 (98)                         | 41 (100)                                           | 70 (97)                                            | 422 (98)                                                 | 1.063<br>p = 0.588                                                                       |                                                                            |                                                                                      |                                                                            |
| Bystander CPR           | 414 (77)<br>n = 541              | 34 (83)                                            | 61 (85)                                            | 319 (75)<br>n = 428                                      | 4.57<br>p = 0.102                                                                        |                                                                            |                                                                                      |                                                                            |
| PAD applied             | 67 (12)                          | 10 (24)                                            | 9 (13)                                             | 48 (11)                                                  | 6.09<br>p = 0.048                                                                        |                                                                            |                                                                                      |                                                                            |
| EMS response, mean      | 4.4 (1.7)                        | 4.7 (1.9)                                          | 4.5 (2.0)                                          | 4.4 (1.6)                                                | 0.56<br>p = 0.571                                                                        |                                                                            |                                                                                      |                                                                            |
| ROSC at end of EMS care | 384 (71)                         | 30 (73)                                            | 43 (60)                                            | 311 (72)                                                 | 4.74<br>p = 0.094                                                                        |                                                                            |                                                                                      |                                                                            |
| Admitted to hospital    | 390 (75)<br>n = 523              | 30 (79)<br>n = 38                                  | 44 (65)<br>n = 68                                  | 316 (76)<br>n = 417                                      | 4.20<br>p = 0.123                                                                        |                                                                            |                                                                                      |                                                                            |
| Discharged alive        | 267 (49)                         | 26 (63)                                            | 37 (51)                                            | 204 (47)                                                 | 4.05<br>p = 0.132                                                                        |                                                                            |                                                                                      |                                                                            |

Values are n (%), unless otherwise indicated. \*n = 544 cases with known location type. <sup>‡</sup>p value calculated using Welch test. <sup>†</sup>p value calculated using Fisher exact test. Abbreviations as in Table 2.

cohort. The lower incidence in traditional exercise facilities in this cohort may be due to a combination of factors. First, the incidence of cardiac arrest has declined over time so that this contemporary cohort may reflect the lower overall SCA incidence. In addition, the current investigation used a much more comprehensive approach to identifying and classifying exercise facilities, which may be reflected in a larger denominator. Nonetheless, the incidence in traditional exercise facilities is substantial and supports the recommendation of AED placement in health clubs. We report a total of 136 arrests at exercise facilities, with substantial occurrence at traditional sites but even more at alternative exercise facilities (52 vs. 84 events, respectively). The site incidence rates of SCA at some of the alternative exercise sites were comparable to the rate at traditional exercise facilities. These results can have implications for AED placement in alternative exercise facilities. In essence, if legislature can mandate AED placement at a traditional exercise facility, a similar strategy would be appropriate for indoor tennis facilities, ice arenas, bowling alleys, and community centers. Conversely, it would likely be resource intensive to place AEDs in all dance studios and martial arts schools because the site incidence rates were low.

This study characterized the activities and locations in a manner that enhances our understanding of SCA that occurs during exercise. The frequency of activities such as basketball, “working out,” and use of a treadmill were not surprising. On the other hand, the high frequency of dancing and bowling, (representing 11.6% and 5.4% of exercise-related arrests, respectively) was unexpected; perhaps this high frequency relates to adverse risk factors and relatively poor fitness level of participants at these sites.

The relatively high occurrence of arrest with swimming, at 4.5% of all reported events, may relate to both the exertion of this exercise and the increased risk of arrest during swimming in individuals with genetic cardiac channelopathies. Long QT syndrome type 1 (LQT1) has long been recognized as being associated with arrest while swimming, and a recent molecular autopsy study demonstrated that nearly 30% of swimming-related “drowning” was associated with LQT1 or another arrhythmogenic channelopathy (12).

In our study cohort of patients experiencing SCA at exercise facilities, three-quarters were participating in high dynamic exercise activities compared with low dynamic exercise, as measured by the Bethesda Conference sports matrix classification.

**Survival of SCA.** Survival of out-of-hospital SCA varies greatly by location. Historically, overall survival rates of <5% have been reported (13–15). Among SCA assessed by EMS, only 8.4% were reported to survive (16). In contrast, in select locations such as aircraft and casinos, survival rates as high as 40% to 53% have been reported, particularly with rapid AED placement (17,18). It is estimated that survival is reduced as much as 5% to 10% for every minute of delay until defibrillation (19,20) so that locations with rapid EMS or placement of an AED would be expected to result in greater rates of survival.

In contrast to SCA at home, where survival rates of 12% do not approach the rate of 34% observed in public locations (21), survival rates at exercise sites appear to exceed survival at other public locations. In the PAD trial, a randomized study of PAD (22), survival was 35% at fitness centers, compared with 18% overall, although this only represented 24 events. We also observed a higher survival to hospital discharge among the patients with SCA at exercise sites (combined traditional and alternative) versus other indoor sites (49% vs. 34%;  $p = 0.001$ ).

**Implications for public health and AED placement.** It is important that communities undertake efforts to improve resuscitation from SCA (23). However, even with recognition of elevated risk for SCA at exercise facilities, the potential benefit of prompt resuscitation with an AED at these sites and the publication of recommendations from expert professional organizations, regulatory and voluntary compliance is incomplete. Our findings should encourage broader implementation of and adherence to recommendations and regulations for AED placement and SCA response protocols at traditional exercise facilities. In addition, these standards should be extended to alternative fitness facilities, where SCA site incidence rates are comparable to those at traditional exercise facilities.

**Study limitations.** This was a retrospective study for which we only examined SCAs that occurred in an indoor public location. As such, we cannot infer results at outdoor exercise facilities or for patients who experienced a cardiac arrest shortly after they left the indoor location where they had been exercising. The study was not designed to include cases that occurred in the setting of exercising at home.

Information regarding denominators for site incidence rates was not available for all location types. Denominators for community centers and hotels with gyms could not be calculated for the year 2002 from the NETS Database; they were estimated based on current available data.

We do not have information as to the number of participants at the exercise facilities, and without this denominator, we could not calculate a frequency of SCA per person in terms of location or the activity in which they were participating. We also do not have data on the age of other participants at the various locations so that no age-related risk could be determined.

The finding of improved survival associated with indoor exercise locations bears consideration. The arrests in our study occurred in patients who were younger and more likely to be male, have a witnessed arrest, have a cardiac etiology, have VF/VT, receive bystander CPR, and have PAD applied. One must also consider characteristics that were not measured. For example, patients at exercise sites may be more fit than the general population, and the very act of going to the exercise site suggests that perhaps they were not feeling ill that day.

Data on PAD application were used to infer PAD availability at the cardiac arrest locations, although we cannot determine if PAD was available and not employed for some

reason. Also, from 1996 to 1998, PAD application was not systematically recorded in Seattle; therefore, PAD application data were only available from data reporting a PAD shock.

Finally, Seattle and King County represent a particular environment with short response time for EMS and high survival rates (16). As such, our data may not be extrapolated to other communities. Alternately, it is possible that even better survival might be observed elsewhere because the placement of the AED at an exercise facility would have a greater relative effect in shortening the time to administration of a shock.

## Conclusions

SCA at exercise facilities was associated with a high rate of survival, likely due to a number of factors related to the physical location and characteristics of the patients. Furthermore, SCA was observed to occur with a relatively high site incidence rate at certain alternative exercise sites as compared with traditional exercise locations. These data have important public health implications for AED placement and the establishment of community approaches to improve resuscitation from SCA.

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**Key Words:** automated external defibrillator ■ cardiopulmonary resuscitation ■ emergency cardiac care ■ sudden cardiac arrest.